ECON 770 Recitation 14

Hypothesis Test and MLE

Nov 13, 2020

THE UNIVERSITY of NORTH CAROLINA at CHAPEL HILL

University of North Carolina at Chapel Hill

Final Exam

- Calm down
- Try all questions that you understand, properly allocate your time
- Make sure you understand most of assignments and lectures
- Write legibly, write legibly, write legibly
- No cheating by UNC Honor Code

Confidence Intervals

- Given $\alpha \in (0,1)$, find k_{1n} and k_{2n} s.t. $P(\theta - k_{1n} \leq \hat{\theta} \leq \theta + k_{2n}) = 1 - \alpha$, then

$$(1-\alpha)\%CI(\hat{\theta}) \equiv [\theta - k_{1n}, \theta + k_{2n}]$$

- E.g.1 (Exact Bands) Suppose that $X_i \sim iid\ N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \sigma^2/n)$. To find a 95% CI ($\alpha = 0.05$) for $\bar{X}$ if we know μ and σ^2, we can obtain k_n by

$$P(\mu - k_n \leq \bar{X} \leq \mu + k_n) = 0.95$$

or equivalently

$$P(-k_n \frac{\sqrt{n}}{\sigma} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq k_n \frac{\sqrt{n}}{\sigma}) = 0.95$$

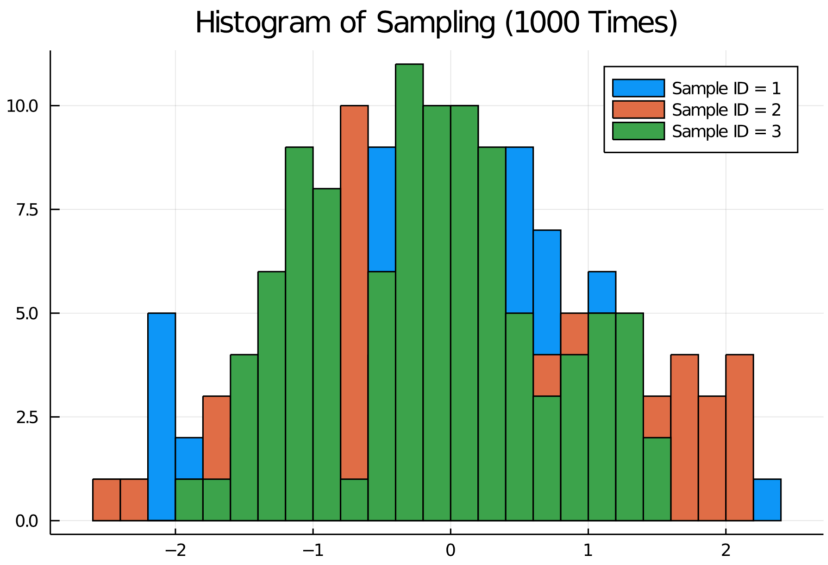
Let $Z_{\alpha/2}$ be the 97.5% quantile of $Z \equiv \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$ and then

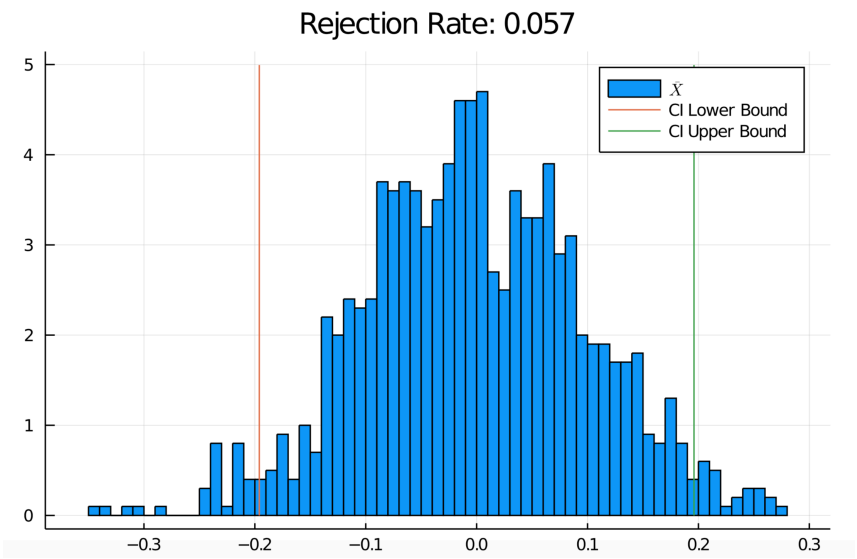
$$k_n \frac{\sqrt{n}}{\sigma} = Z_{\alpha/2} \quad \Rightarrow \quad k_n = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \Rightarrow \quad CI = [\mu - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \mu + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}]$$

CI Illustration

$$P(-k_n \frac{\sqrt{n}}{\sigma} \leq \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \leq k_n \frac{\sqrt{n}}{\sigma}) = 0.95$$

$$k_n \frac{\sqrt{n}}{\sigma} = Z_{\alpha/2} \quad \Rightarrow \quad k_n = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \Rightarrow \quad CI(\bar{X}) = [\mu - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \mu + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}]$$





Confidence Intervals (Contd)

- E.g.2 (Asymptotic Bands) Suppose that $X_i \sim iid\ (\mu, \sigma^2)$. If we know μ, σ^2 and distribution of $\bar{X}$, we can obtain k_n or $(1-\alpha)$%CI for $\bar{X}$ by

$$P(\mu - k_n \leq \bar{X} \leq \mu + k_n) = 1 - \alpha$$

Since CLT tells us $Z_n \equiv \sqrt{n}\left(\frac{\bar{X}-\mu}{\sigma}\right) \xrightarrow{d} N(0,1)$, thus when $n \longrightarrow \infty$

$$P(-k_n\frac{\sqrt{n}}{\sigma} \leq \sqrt{n}\left(\frac{\bar{X}-\mu}{\sigma}\right) \leq k_n\frac{\sqrt{n}}{\sigma}) \ \longrightarrow\ P(-Z_{\alpha/2} \leq Z \leq Z_{\alpha/2})$$

where $Z_{\alpha/2}$ be the $(1-\frac{\alpha}{2})$% quantile of $Z \sim N(0,1)$. and then

$$k_n\frac{\sqrt{n}}{\sigma} = Z_{\alpha/2} \quad \Rightarrow \quad k_n = Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \quad \Rightarrow \quad CI = [\mu - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \mu + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$$

Confidence Intervals (Exercises)

Suppose that $\{X_i\}_{i=1}^n \sim i.i.d\ N(\mu, \sigma^2)$ where μ is unknown, let $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$ and $s_x^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2$, then

- If σ^2 is known, then

$$(1-\alpha)\%CI(\mu) = \left[\bar{X} - \frac{\sigma}{\sqrt{n}} Z_{\alpha/2}, \bar{X} + \frac{\sigma}{\sqrt{n}} Z_{\alpha/2}\right]$$

- If σ^2 is unknown, then

$$(1-\alpha)\%CI(\mu) = \left[\bar{X} - \frac{s_x}{\sqrt{n}} t_{n-1}(\alpha/2), \bar{X} + \frac{s_x}{\sqrt{n}} t_{n-1}(\alpha/2)\right]$$

where $Z_{\alpha/2}$ is the $(1-\alpha/2)$-th quantile of $N(0,1)$, $t_{n-1}(\alpha/2)$ is the $(1-\alpha/2)$-th quantile of t_{n-1}.

Confidence Intervals (Exercises)

Suppose that $\{X_i\}_{i=1}^n \sim i.i.d\ N(\mu, \sigma^2)$ where σ^2 is unknown, let $\tilde{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \mu)^2$ and $s_x^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2$, then

- If μ is known, then

$$(1-\alpha)\%CI(\sigma^2) = \left[\frac{n\tilde{\sigma}^2}{\chi_n^2(\alpha/2)}, \frac{n\tilde{\sigma}^2}{\chi_n^2(1-\alpha/2)}\right]$$

- If μ is unknown, then

$$(1-\alpha)\%CI(\sigma^2) = \left[\frac{(n-1)s_x^2}{\chi_{n-1}^2(\alpha/2)}, \frac{(n-1)s_x^2}{\chi_{n-1}^2(1-\alpha/2)}\right]$$

where $X_n^2(c)$ is the c-th quantile of χ_n^2

Confidence Intervals (Extension)*

Suppose that $\{X_i\}_{i=1}^n \sim iid\ N(a, \sigma_1^2)$, $\{Y_i\}_{i=1}^m \sim iid\ N(b, \sigma_2^2)$. Also suppose independent $\{X_i\}_{i=1}^n$ and $\{Y_i\}_{i=1}^m$. Define

$$S_X^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2, \quad S_Y^2 = \frac{1}{m-1}\sum_{i=1}^{m}(Y_i - \bar{Y})^2$$

- How to construct the CI for $(b - a)$ if $m = n$?
- What if $m \neq n$ but σ_1^2, σ_2^2 are given?
- What if $m \neq n$, σ_1^2, σ_2^2 are unknown but $\sigma_1^2 = \sigma_1^2$?

Hypothesis Test

- Some definitions: Null hypothesis H_0, Alternative hypothesis H_1

$$H_0 : \beta_k = b_k, \quad H_1 : \beta_k \neq b_k, \quad \text{or} \quad H_1 : \beta_k > b_k$$

$$H_0 : \beta \in B_0 \subset \mathbb{R}^K, \quad H_1 : \beta \notin B_0$$

- Binary decision rule $d \in \{0, 1\}$:

$$d : (X_1, \ldots, X_k) \to \{0, 1\}$$

  where $d = 1$ means reject H_0 and $d = 0$ means do not reject H_0
- Rejection (critical) region: R_c s.t. $d = 1$
- Test statistics $T(X) : \mathbb{R}^k \to \mathbb{R}$
- Type I Error / Size of the test / Prob(reject|H_0 is true): α
  Type II Error: γ; Power of the test: $1 - \gamma$

	H_0	H_1
$d = 0$	$1 - \alpha$	γ
$d = 1$	α	$1 - \gamma$

Hypothesis Test (Contd)

- Let $X_i \sim iid\ (\mu, \sigma^2)$, we know that by CLT,

$$\sqrt{n}\left(\frac{\bar{X}-\mu}{s_x}\right) \xrightarrow{d} N(0,1)$$

  where $\sqrt{n}\left(\frac{\bar{X}-\mu}{s_x}\right)$ is exactly t_{n-1} if $X_i \sim iid\ N(\mu, \sigma^2)$
- Define $H_0 : \mu = \mu_0$ (v.s. $H_1 : \mu \neq \mu_0$ or $H_1 : \mu > \mu_0$)
- Idea of t-test: if H_0 is true, then data will show us

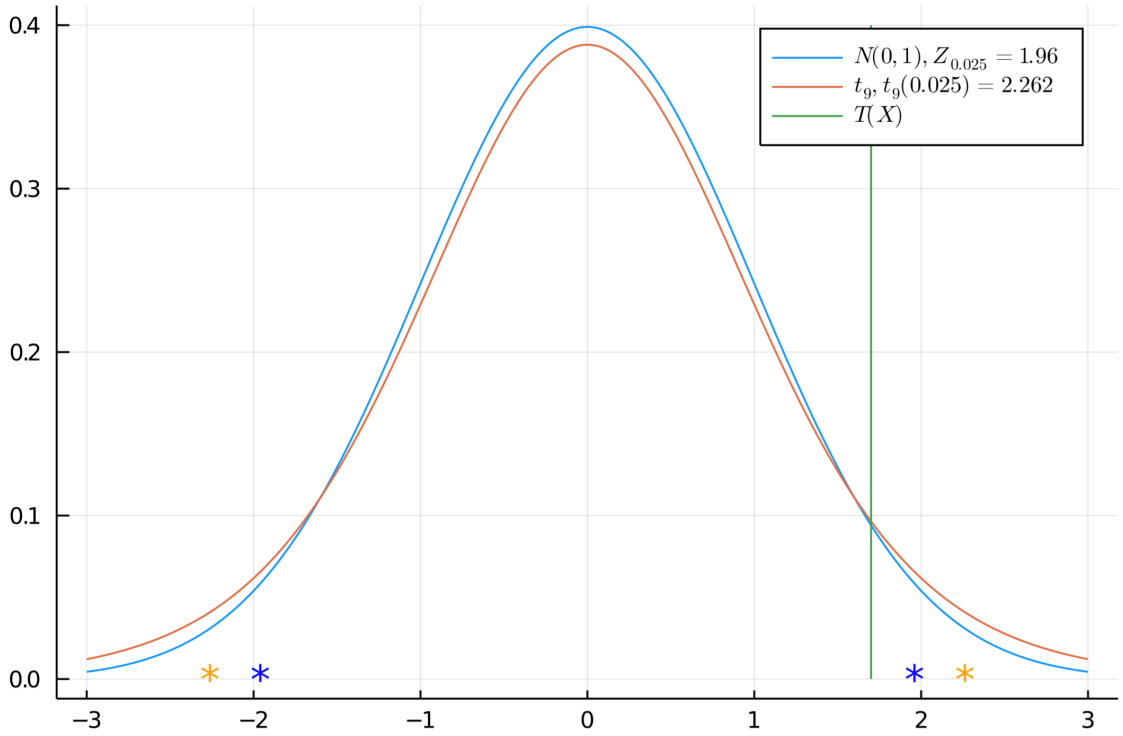
$$T \equiv \sqrt{n}\left(\frac{\bar{X}-\mu_0}{s_x}\right) \xrightarrow{d,H_0} N(0,1)$$

  or if $X_i \sim iid\ N(\mu, \sigma^2)$,

$$T \equiv \sqrt{n}\left(\frac{\bar{X}-\mu_0}{s_x}\right) \overset{H_0}{\sim} t_{n-1}$$

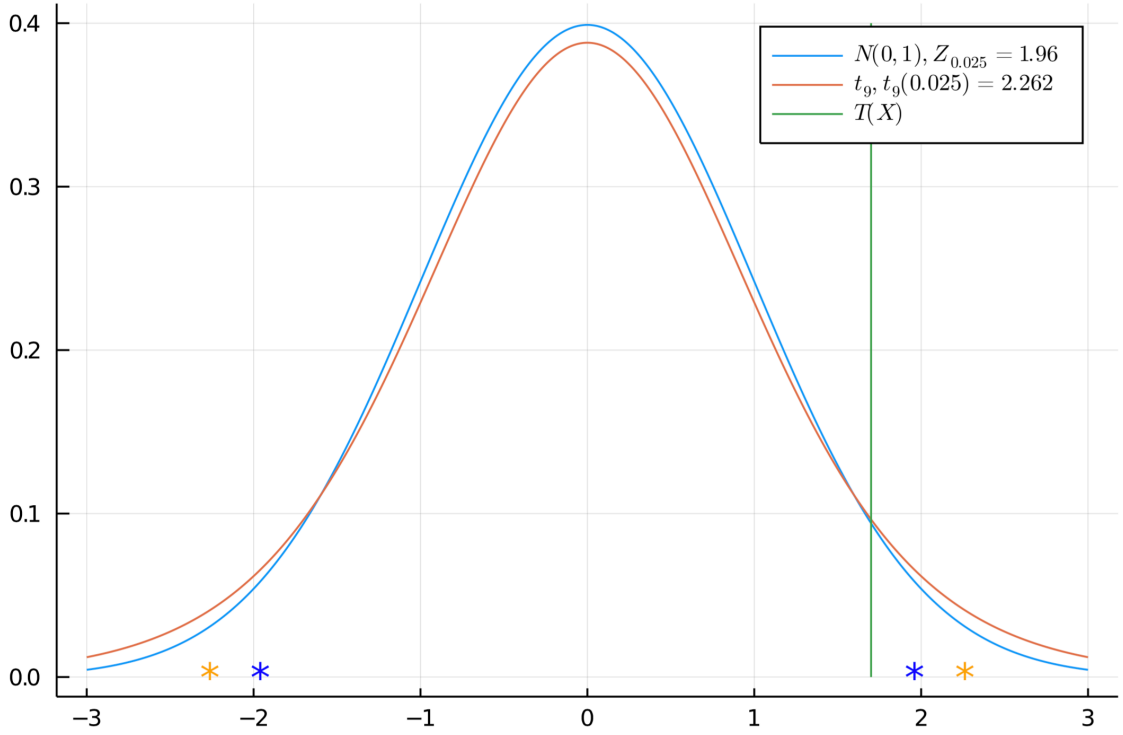
Critical Value and P-value (t-test)

- Critical value $c \in \mathbb{R}$ s.t.
  - One-sided: $\alpha = P\left(\frac{\bar{X}-\mu_0}{s_x/\sqrt{n}} > c|H_0\right) \Rightarrow c = t_{n-1}(\alpha)$ or $c = Z_\alpha$
  - Two-sided: $\alpha = P\left(\left|\frac{\bar{X}-\mu_0}{s_x/\sqrt{n}}\right| > c|H_0\right) \Rightarrow c = t_{n-1}(\alpha/2)$ or $c = Z_{\alpha/2}$
- P-value:
  - t-dist, one-sided: $p = P(t_{n-1} > T)$ (right tail) or $p = P(t_{n-1} < -T)$ (left tail)
  - t-dist, two-sided: $p = 2P(t_{n-1} > |T|)$



Critical Value and P-value (t-test, Contd)

- Reject if $T \in R_c = (c, \infty)$ or $T \in R_c = (-\infty, -c) \cup (c, \infty)$
  Reject if p-value $< \alpha$



Power Function

- Suppose $H_0 : \mu = \mu_0$ v.s. $H_1 : \mu > \mu_0$
- Choose a point alternative $\tilde{\mu}$ s.t. $H_1 : \mu = \tilde{\mu} > \mu_0$
- Power

$$P_n(\tilde{\mu}) \equiv P\left(\frac{\bar{X} - \mu_0}{s_x/\sqrt{n}} > c|\mu = \tilde{\mu}\right) = P(\text{reject } H_0|H_1)$$

  which is the power against $H_1 : \mu = \tilde{\mu} > \mu_0$
- If μ_0 in H_0 is very different from $\tilde{\mu}$ which is true, then by

$$P\left(\frac{\bar{X} - \mu_0}{s_x/\sqrt{n}} > c|\mu = \tilde{\mu}\right) = P\left(\frac{\bar{X} - \tilde{\mu}}{s_x/\sqrt{n}} + \frac{\tilde{\mu} - \mu_0}{s_x/\sqrt{n}} > c|\mu = \tilde{\mu}\right)$$

  we will observe large $P_n(\tilde{\mu})$

Neyman-Pearson Lemma (LR-test, 1933)

- Likelihood function $L(\theta|X)$ is the joint p.m.f or p.d.f where $\theta \in \mathbb{R}^k$ is a (vector) distribution parameter and X is the data

$$L(\theta|X) = f_n(x_1, \ldots, x_n|\theta)$$

- Define $H_0 : \theta = \theta^0$ vs $H_1 : \theta = \theta^1$
- The most powerful[1] (highest power) test at level α is to reject H_0 if

$$\text{Likelihood Ratio} \quad \Lambda(X) \equiv \frac{L(\theta^1|X)}{L(\theta^0|X)} > c$$

  where c is given by $P(\Lambda(X) > c|H_0) = \alpha$
- For fixed level $Pr(\textit{Type I Error}) = \alpha$, LR test has the lowest $Pr(\textit{Type II Error}) = \gamma$
- Example: LR Test of mean of Gaussian distribution

[1] also called admissible at level α

Maximal Likelihood Estimation

- Suppose that i.i.d sample $X_i \sim F(\theta_0)$ where $\theta_0 \in \mathbb{R}^k$ is the true distribution parameter
- Sample likelihood function

$$L_n(\theta) = f(x_1, \ldots, x_n|\theta) = \prod_{i=1}^{n} f(x_i|\theta)$$

- Choose $\hat{\theta}$ to make data $x_1, \ldots, x_n$ most likely and we get maximum likelihood estimator (MLE) on compact $\Theta \subset \mathbb{R}^k$

$$\hat{\theta}_n = \arg\max_{\theta \in \Theta} L_n(\theta)$$

- In practice we do a monotonic transformation and deal with log-likelihood

$$l(\theta) = \mathbb{E}\left[\ln(f(X_i|\theta))\right], \quad l_n(\theta) = \ln(L_n(\theta))$$

$$\hat{\theta}_n = \arg\max_{\theta \in \Theta} l_n(\theta) = \arg\max_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} \ln f(x_i|\theta)$$

Consistency of MLE

Assumption (Identification)

The parameter of interest θ_0 is unique in the sense that for all $\theta \neq \theta_0$,

$$l(\theta_0) > l(\theta)$$

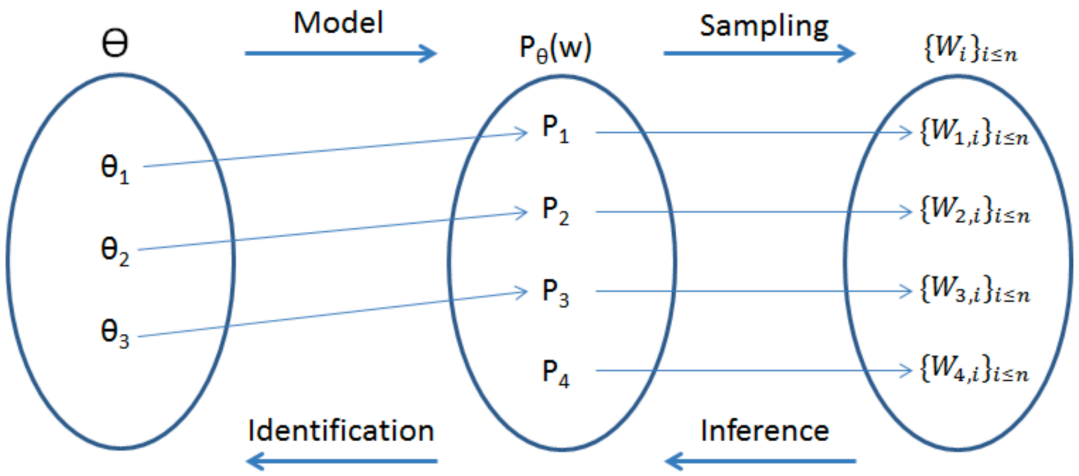


Figure: Duke Econ 883.006 SlidesH4 (Prof. Bugni, 2020)

Consistency of MLE (Contd)

Theorem

If $X \sim F(\theta_0)$ with pdf $f(x|\theta_0)$ is true for unique θ_0 in the interior of Θ, then

$$\theta_0 = \arg\max_{\theta \in \Theta} l(\theta) = \arg\max_{\theta \in \Theta} \mathbb{E}\left[\ln f(X_i|\theta)\right]$$

Lemma (In the proof in lecture slides)

If

$$f(x|\theta) \overset{a.s.}{=} f(x|\theta_0) \quad \Rightarrow \quad \theta = \theta_0$$

then

$$l(\theta_0) > l(\theta) \quad \forall\, \theta \neq \theta_0$$

Theorem (Invariance)

If $\hat{\theta}_n$ is the MLE of θ_0, then for any measurable[a] function $h(\cdot) : \mathbb{R}^k \to \mathbb{R}^q$, $\hat{\beta}_n = h(\hat{\theta}_n)$ is the MLE of $\beta_0 = h(\theta_0)$

[a] Not hard to prove if assume bijection $h(\cdot)$

Consistency of MLE (Contd)

Assumption (ULLN)

- $\ln f(X_i|\theta)$ is continuous and measurable for all $\theta \in \Theta$
- $\mathbb{E}\left[\sup_{\theta\in\Theta} |\ln f(X_i|\theta)|\right] < \infty$
- Θ is compact

Theorem (Consistency)

Given *Identification* and *ULLN* assumptions,

$$\hat{\theta}_n \xrightarrow{p} \theta \quad (\text{Actually } \hat{\theta}_n \xrightarrow{a.s.} \theta)$$

MLE - Some Definitions

Let $l_n(\theta) = \sum_{i=1}^{n} \ln f(X_i|\theta)$, then

- Score/Gradient $S_n(\theta) = \dfrac{\partial}{\partial\theta} l_n(\theta) = \sum_{i=1}^{n} \dfrac{1}{f(X_i|\theta)} \dfrac{\partial}{\partial\theta} f(X_i|\theta)$
- Theorem 1: If $X_i \sim f(x|\theta_0)$ for all i, then

$$\mathbb{E}\left[S_n(\theta_0)\right] = 0$$

- Theorem 2: Let $X_i \sim$ *iid* $f(x|\theta_0)$, if $\left\|E\left[\frac{\partial}{\partial\theta} \ln f\left(X_i \mid \theta_0\right) \frac{\partial}{\partial\theta} \ln f\left(X_i \mid \theta_0\right)'\right]\right\| < \infty$, then

$$\frac{1}{n} S_n(\theta_0) \xrightarrow{p} 0$$

- Hessian Matrix: $H_n(\theta) = -\dfrac{\partial^2}{\partial\theta\partial\theta'} l_n(\theta) = -\sum_{i=1}^{n} \frac{\partial^2}{\partial\theta\partial\theta'} \ln\left(f\left(X_i \mid \theta\right)\right)$
- Expected Hessian $H(\theta) = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta\partial\theta'} \ln\left(f\left(X_i \mid \theta\right)\right)\right]$
- Fischer Information Matrix $I_\theta = \mathbb{E}\left[SS'\right] =$
$\mathbb{E}\left[\frac{1}{f(X_i|\theta_0)^2}\left(\frac{\partial}{\partial\theta} f\left(X_i \mid \theta_0\right)\left(\frac{\partial}{\partial\theta} f\left(X_i \mid \theta_0\right)\right)'\right]\right. = \mathbb{E}\left[\frac{1}{f(X_i|\theta_0)^2} \frac{\partial^2}{\partial\theta\partial\theta'} f\left(X_i \mid \theta_0\right)\right]$

Cramér-Rao Lower Bound

Theorem

If $\tilde{\theta}_n$ is an unbiased estimator of θ_0, then

$$Var[\tilde{\theta}_n] \geq \frac{1}{n} I_\theta^{-1} \equiv \text{Cramér-Rao Lower Bound}$$

where $I_\theta = \mathbb{E}[SS'] = \mathbb{E}\left[\frac{1}{f(X_i|\theta_0)^2} \frac{\partial^2}{\partial\theta\partial\theta'} f(X_i \mid \theta_0)\right]$

Asymptotic Normality

Theorem

Under $X_i \sim$ *iid* $f(x|\theta_0)$ and some regularity conditions,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I_\theta^{-1})$$

- Proved by ULLN, Taylor Expansion, CLT and Slutsky Theorem
- MLE is asymptotically efficient
- MLE is asymptotically unbiased and consistent